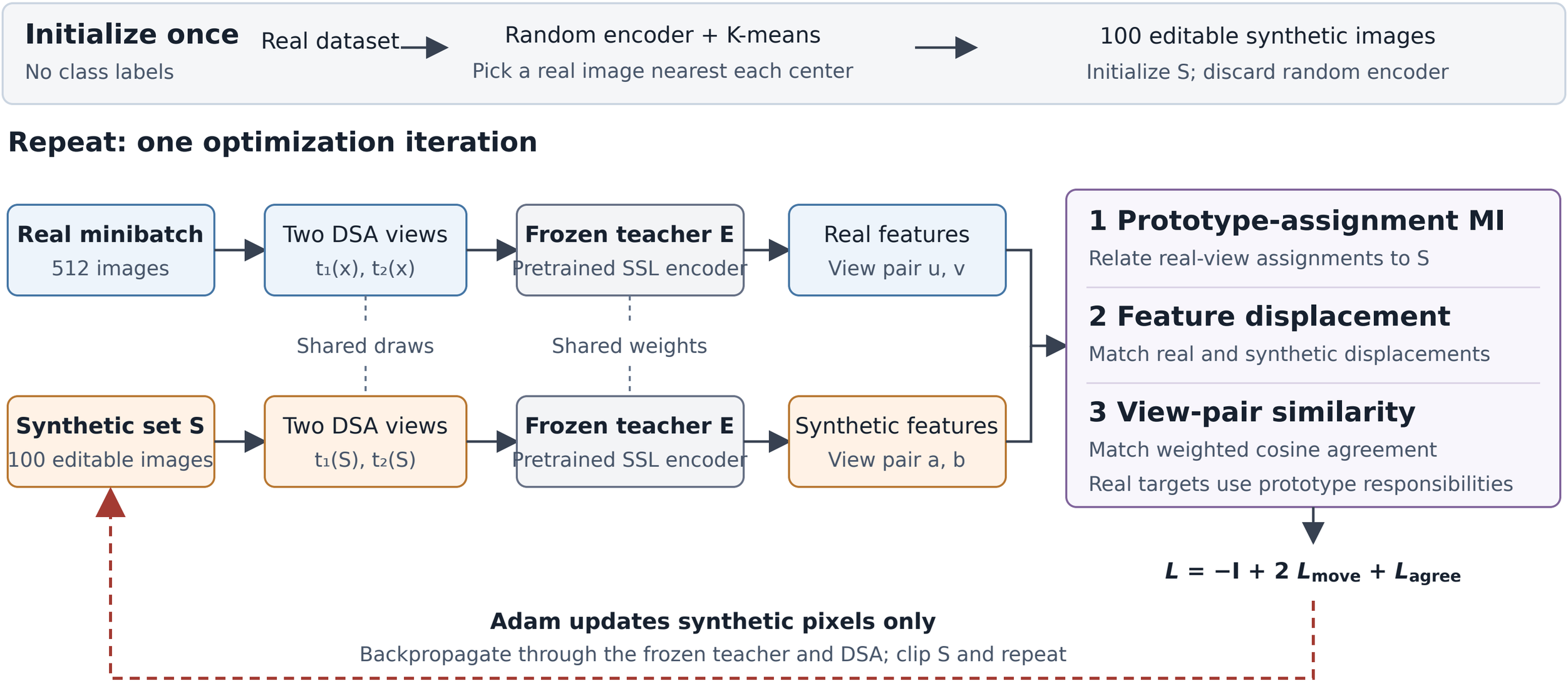
Initialize once
No class labels
Real dataset
Random encoder + K-means
Pick a real image nearest each center
100 editable synthetic images
Initialize S; discard random encoder
Repeat: one optimization iteration
Real minibatch
512 images
Two DSA views
t₁(x), t₂(x)
Frozen teacher E
Pretrained SSL encoder
Real features
View pair u, v
Shared draws
Shared weights
Synthetic set S
100 editable images
Two DSA views
t₁(S), t₂(S)
Frozen teacher E
Pretrained SSL encoder
Synthetic features
View pair a, b
1 Prototype-assignment MI
Relate real-view assignments to S
2 Feature displacement
Match real and synthetic displacements
3 View-pair similarity
Match weighted cosine agreement
Real targets use prototype responsibilities
L = −I + 2 L_move + L_agree
Adam updates synthetic pixels only
Backpropagate through the frozen teacher and DSA; clip S and repeat